

ENGINEERING · DELIVERY PLAN

Build Timeline

Danyal → 24 Jul · Haseeb → 29 Jul

A weekday-by-weekday, module-by-module plan for both AIOS builds — Danyal now spans **five modules** (Audit · Content · Off-page · Portal · Policy Radar) on a Google-Sheets store — owned by Adan, Huzaifa and Arham, with the critical path, integration gates, cross-system reuse, and an honest capacity read.

Adan

Backend · Infra · AI · Lead

Huzaifa

Frontend · Portals · UI

Arham

Database · RLS · Jobs

24 & 29 Jul

Two ships, 5 days apart

PREPARED FOR

Engineering
Adan · Huzaifa · Arham

PREPARED BY

Xegents

DATE

9 July 2026

WINDOW

Weekdays
9 → 29 Jul

01

GROUND RULES

The plan at a glance

2

SYSTEMS IN FLIGHT
(SEPARATE REPOS)

3

DEVELOPERS, ROLE-SPLIT
BY LAYER

12

WEEKDAYS TO DANYAL
(9→24 JUL)

15

WEEKDAYS TO HASEEB
(9→29 JUL)

THE WORKING CALENDAR — WEEKDAYS ONLY, ~8H/DAY

WK 0		WEEK 1					WEEK 2					WEEK 3		
9 THU	10 FRI	13 MON	14 TUE	15 WED	16 THU	17 FRI	20 MON	21 TUE	22 WED	23 THU	24 FRI ↕D	27 MON	28 TUE	29 WED ↕H

SHIP CRITERION — DANYAL (FRI 24 JUL)

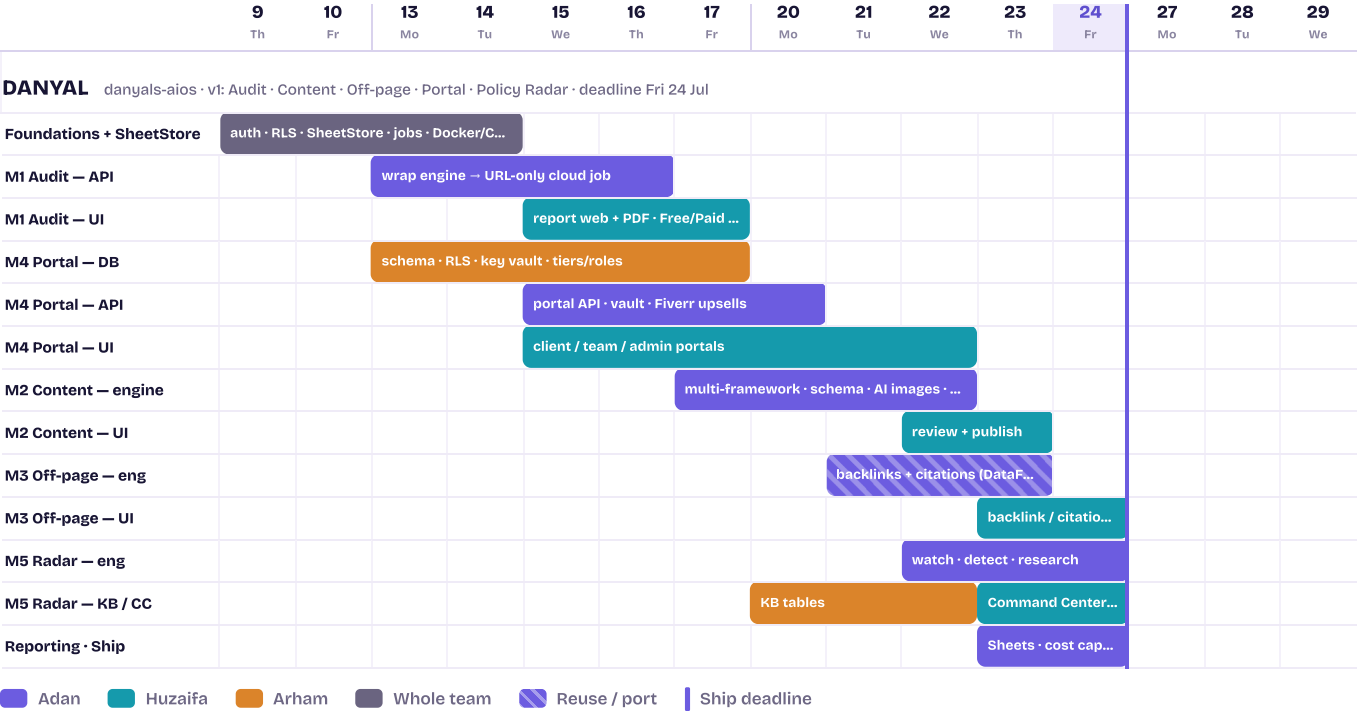
All five v1 modules live on Danyal's VPS: URL-only Audit (existing engine), multi-framework Content (AIDA/PAS/... → schema → AI images → WordPress/PDF, review gate), the new Off-page engine (backlinks · citations · Web 2.0 via official APIs), the four-role Portal (Fiverr upsells, key vault, Free/Paid tiers), and Policy Radar feeding the Command Center — all on the Sheets-as-store spine with cost caps. **Free = audit-only; Paid unlocks the rest.** A human approves before anything reaches a client.

SHIP CRITERION — HASEEB (WED 29 JUL)

All 15 clients live with the P0 core + P1 pulled in: dual-mode Ingestion (GSC/GA4 + DataForSEO + CSV), nightly Rank Tracking, freshness-gated branded Reports — then Technical Audit, Backlink analysis, Task board and Keyword Research as the P1 tier. Deployed under Haseeb's domain, used for real.

Danyal — 12 weekdays

The **danyals-aos** track. Bars are coloured by owner and sized to weekday span; the violet line is the **24 Jul** ship. Foundations carry the **SheetStore** spine, then the **five** modules run overlapped — Portal is the long pole, Off-page (striped) reuses the audit's DataForSEO client, and Policy Radar lands last.

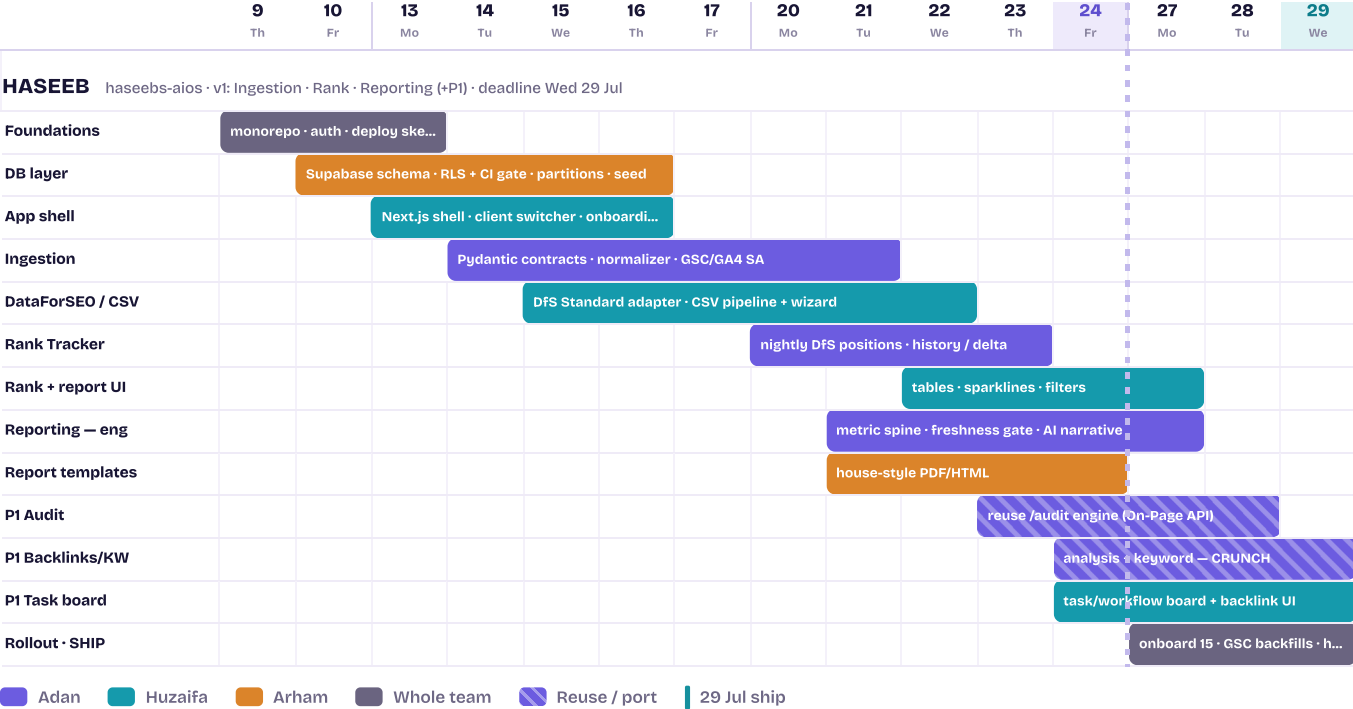


CRITICAL PATH

Foundations + SheetStore → **M4 Portal** (widest surface: three role views + Free/Paid tiers + vault) → Off-page and Policy Radar close the loop. M1 Audit is **URL-only** and wraps the existing engine, so it lands early and de-risks the demo. The tail piles onto Adan — Content, Off-page and Radar overlap in the final week; that crunch is what the cut-list protects.

Haseeb — 15 weekdays

The **haseebbs-aos** track. Same team, so this runs behind Danyal: the dashed line marks Danyal's 24 Jul ship, the teal line Haseeb's **29 Jul**. Ingestion is built first (it's the hardest and everything inherits it); the striped P1 bars reuse Danyal's engine + patterns and are the crunch zone.



The linchpin is split, not single-owned

Adan freezes the canonical Pydantic contracts + owns the normalizer; Huzaifa wires the DataForSEO adapter and CSV pipeline against them. One integration deep (GSC + DataForSEO Standard + CSV), not five shallow — so a lost day doesn't sink Weeks 3–4. CSV is the primary onboarding path; a client ships the moment GSC-or-CSV + seed keywords land.

04 DANYAL · MODULE BY MODULE

Danyal — build detail

MODULE	KEY TASKS	OWNER	TARGET
Foundations + SheetStore	Auth + roles, RLS/tenancy wall, Redis + Celery queues, Docker Compose + Caddy TLS, Next.js app shell + client switcher, encrypted key-vault scaffold, CI — plus the SheetStore adapter so operational data lands in per-client Google Sheets (Postgres keeps identity, secrets & the Radar KB).	Adan Arham Huz	Mon 14 Jul
M1 · Audit ♦ URL-only	Wrap the existing audit_engine as a Celery job + API; URL-only — no logins (GSC/GA4 are optional owner-authorized enrichers); findings → client Sheet + 20–30p house PDF; portal Run audit with the Free/Paid tier gate; report web + PDF view.	Adan Huz	Thu 16 Jul
M2 · Content	Multi-framework (AIDA · PAS · BAB · FAB · 4 Ps · PASTOR · 4 U's) auto-selected by type + intent → JSON-LD schema → AI images + alt → WordPress REST publish + manual-PDF path; human review checkpoint; per-job cost logging.	Adan Huz	Wed 22 Jul
M3 · Off-page ♦ new in v1	Backlink + citation monitoring (DataForSEO Business Data / CSV → new/lost alerts) and Web 2.0 publishing via WordPress.com / Blogger / Tumblr official APIs → verify live → track; human quality gate + footprint diversification. Paid tier ; records to Sheets.	Adan Huz	Thu 23 Jul
M4 · Portal ♦ long pole	Client / team / admin portals; dashboard + reports (web+PDF) + auto-milestones; Fiverr upsell cards; key-vault admin; Free/Paid tiers + role provisioning. Manager view optional.	Huz Adan Arham	Wed 22 Jul
M5 · Policy Radar	Scheduler + source change-detection; research → flagged KB (severity / category / region) in Postgres; Command Center recommendations; human-confirm gate; closed loop into audit checks.	Adan Arham Huz	Thu 23 Jul
Reporting · Ship	Per-client Google-Sheets workbooks (service account, quota-safe batched writes); per-client spend caps + daily kill-switch; security pass; nightly backups; UAT with Danyal; production deploy + handover.	Adan Team	Fri 24 Jul ♦

Reuse, not rebuild

M1 is the tested danyals-audit-system engine wrapped as a job; Off-page's backlink/citation data reuses the same **DataForSEO** client — the fastest safe path and why the demo lands in Week 1.

Freeze contracts early

Adan freezes the Portal + Content + SheetStore shapes by Jul 15 so Huzaifa's UI and Arham's schema build against a fixed target — no moving interface mid-week.

05 HASEEB · MODULE BY MODULE · PO

Haseeb — foundations & core

MODULE	KEY TASKS	OWNER	TARGET
Foundations	Monorepo (/api,/web); Supabase project; core schema + FORCE RLS + CI gate ; pg_cron partition-maker + DEFAULT partition; credential envelope-encryption (AES-GCM, off-DB key); Supabase Auth → Next middleware; Docker/Caddy deploy skeleton; app shell + onboarding form v0.	Arham Adan Huz	Tue 15 Jul
Ingestion (11) ♦ hardest	Freeze canonical Pydantic contracts (RankingRow, BacklinkRow); GSC SA paginated pull (25k/req); GA4 SA staggered pull + 429-backoff; DataForSEO Standard (task-POST → pingback → collect); cost-logged; CSV/XLSX pipeline (upload→sanitize→map→validate→commit); normalizer; mapping-wizard UI; SEMrush/Ahrefs templates.	Adan Huz Arham	Tue 21 Jul
Rank Tracker (4)	Keyword lists; nightly DataForSEO Standard positions; history + delta; freshness stamp; rank UI (tables, sparklines, filters). DataForSEO SERP is the only rank source — GSC is visibility only.	Adan Huz	Thu 23 Jul
Reporting (9)	metric_daily rollup + freshness gate ; per-client house-style PDF/HTML (traffic + rankings + backlinks); AI narrative (Sonnet 5, batch, json_schema) → draft → approved gate; pg_cron scheduling.	Adan Arham Huz	Mon 27 Jul

WHY THIS ORDER

Ingestion is built **first and early** because every module inherits it and it's the highest-risk integration. Rank + Reporting then turn free GSC/GA4 + cheap DataForSEO into the billable, branded deliverable that makes AIOS *fee/* like a product — the PO three carry the ship; everything after is upside.

Haseeb — P1 & rollout

MODULE	KEY TASKS	OWNER	TARGET
P1 · Technical Audit	On-Page API → pre-summarized findings → /audit service (hard turn-cap); reuses the haseeb-ai-audit engine.	Adan	Mon 27 Jul
P1 · Backlinks	Analysis / monitor only — inventory, new/lost diff, domain-strength & toxic score. Read-only; no outreach (Phase 3).	Adan Huz	Tue 28 Jul
P1 · Task Board	Create / assign / track tasks; audit & alert findings land here; board UI.	Huz Arham	Tue 28 Jul
P1 · Keyword Research	Volume / difficulty ingest + grouping; assign keywords to clients & pages.	Adan	Wed 29 Jul
Rollout · Ship	Onboard all 15 (GSC/GA4 SA or CSV; seed keywords; throttled backfills over days); scheduling live; report polish; hardening (RLS/GUC audit, spend dashboard + kill-switch test); docs (user guide + import runbook); UAT with Haseeb.	Team	Wed 29 Jul ♦

CROSS-SYSTEM REUSE — WHY THE SECOND BUILD IS A PORT, NOT A REBUILD

What Haseeb inherits from Danyal

Frozen Pydantic contracts, the auth + FORCE-RLS pattern + CI gate, the Docker/Caddy deploy skeleton, the single ai/claude.py wrapper, and the house PDF/report stack. Built once on Danyal, ported to the second repo.

WHAT'S GENUINELY NET-NEW FOR HASEEB

The dual-mode data layer with no Danyal equivalent: **DataForSEO Standard**, GSC/GA4 service-account pulls, the CSV/XLSX normalization pipeline, and the 15-client agency ops. That delta is the real Haseeb work — the striped P1 bars.

Who owns what, across both

Adan

BACKEND · INFRA · AI · LEAD

- Both: FastAPI core, routers/services, Celery workers, the ai/claude.py wrapper, deploy + CI, spend caps & kill-switch, review gate.
- Danyal: URL-only audit wrap, multi-framework Content, **Off-page** (DataForSEO backlinks/citations + Web 2.0 publish), Policy Radar, SheetStore + Sheets reporting.
- Haseeb: frozen contracts + normalizer, GSC/GA4, DataForSEO Standard, rank + reporting engine, AI narrative.

Load ~90% · the critical path runs through here — protect it.

Huzaifa

FRONTEND · PORTALS

- Both: Next.js 15 app shell, TanStack Query wiring, all portal/dashboard UI, Recharts.
- Danyal: client/team/admin portals, audit report view, content review UI, **backlink / citation + Web 2.0 tracker**, Command Center UI.
- Haseeb: onboarding + client switcher, CSV mapping wizard, rank + report UI, backlink/task-board UI.

Builds against Adan's frozen API contracts — never a moving target.

Arham

DATABASE · RLS · JOBS

- Both: schema + migrations, **FORCE RLS + CI gate**, pg_cron partitions, seed data, house report templates + QA.
- Danyal: portal/vault/**tiers** schema, SheetStore workbook templates, Policy-Radar KB tables (severity/region), seed clients.
- Haseeb: tenancy + ranking partition schema, RLS policies, import templates, report templates, data QA.

Owns the tenant wall — no table ships without a policy (CI-enforced).

CADENCE — THE SAME ON BOTH TRACKS

Daily

15-min async standup — blockers first.

Monday

Plan the week off this board.

Wednesday

Integration check — merge, not diverge.

Friday

Demo-or-die on staging.

08 MILESTONE GATES

Prove it every Friday

Each gate is a concrete, demoable exit — not a status colour. If a Friday gate misses, the cut-list (next page) triggers on Monday, not on ship day.

DANYAL

14
JUL**G1 · Foundations + SheetStore**

Login → create client → switcher; RLS gate passing; SheetStore writes an audit's findings to a client Sheet; a URL-only audit returns a PDF.

18
JUL**G2 · Audit + Portal shell**

A provisioned client logs in, runs a **URL-only** audit on the **Free** tier, downloads the report.

22
JUL**G3 · Content + Off-page + Radar**

Multi-framework generate → review → publish to WordPress; **Paid** unlocks Off-page — first backlinks/citations land in the Sheet; Command Center shows a live rec.

24
JUL**◆ SHIP**

Full Free/Paid client journey + Off-page + admin oversight live on Danyal's VPS; UAT signed off.

HASEEB

15
JUL**M1 · Foundations + RLS**

Login → client → switcher on staging; RLS CI gate green; partition auto-create proven; 15 clients seeded.

21
JUL**M2 · Ingestion works**

Real Ahrefs CSV → normalized rows; DataForSEO rank tasks pingback-collect for ≥2 pilots; GSC nightly for ≥5.

24
JUL**M3 · Report end-to-end**

One pilot: ingestion → rank → freshness-gated report, generated end-to-end. P1 pulled in only if solid.

29
JUL**◆ SHIP**

15 clients live (some manual-mode), nightly jobs running, reports generating, agency trained.

Capacity, risk & the cut-list

THE CAPACITY REALITY

Two full v1 platforms, **3 developers, weekdays only** — 12 working days to Danyal, 15 to Haseeb, both off the **same** team. And Danyal's v1 just grew from four modules to **five: Off-page** (backlinks, citations, Web 2.0) is now in scope on the same 12-day window. That is roughly one senior FTE-month of effective capacity against what the standalone plans each scope at 3–5 weeks. It only closes because the second build **ports** the first and Off-page **reuses** the audit's DataForSEO client. Treat Danyal as the reference implementation, its PO (Audit · Content · Portal · Radar) as mandatory core, and **Off-page's Web 2.0 as the first thing the cut-list protects**.

TOP RISKS & THE GUARDRAIL

RISK	GUARDRAIL
Missing RLS on a new table → tenant leak	FORCE RLS + Week-1 CI gate + service-role GUC wall
Off-page added late → Adan's tail overloads	Reuse the audit's DataForSEO client; Web 2.0 is cut-list-first
Sheets API quota during bulk writes	Redis write-buffer + batched batchUpdate; PG for joins/secrets
Web 2.0 publishing reads as link spam	Human quality gate + genuine articles + footprint diversification
API cost runaway	Per-client caps + org daily kill-switch (DataForSEO + Claude)
Two ships, one team, 5 days apart	Danyal ships first; Haseeb reuses its spine; features freeze 1 day pre-ship

CUT-LIST — IF BEHIND, DROP IN THIS ORDER

- **Off-page Web 2.0 first** (Danyal): the WP.com/Blogger/Tumblr publishing pipeline defers to August; keep backlink/citation **monitoring** only.
- **Haseeb P1 tier**: Keyword Research → Task board → Backlinks → Technical Audit move to August wholesale.
- **GA4** — keep GSC (the SEO signal); traffic becomes a manual screenshot in v1.
- **Scheduled delivery** → on-demand generate + manual send.
- **Danyal Policy Radar** → ship baseline recommendations only; defer change-detection automation.
- **Manager portal** (Danyal) folds into Admin; multi-user roles → single shared org for July.

The rule: the core three (Haseeb) and the core four (Danyal: Audit · Content · Portal · Radar) carry ship. Off-page and everything else is negotiable before either deadline slips.

THE ONE LINE TO REMEMBER

Free proves value, Paid runs the platform. Danyal is the reference build; Haseeb is the port. Ship the core, protect Adan's critical path, cut Web 2.0 first.

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